



## **Final Report**

### **Volatility Forecast and Analysis of Cryptocurrencies Across Different Frequencies**

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## Introduction

Understanding the volatility of assets is crucial for portfolio and asset management as it allows traders to make informed trades. While predicting prices of financial assets like stocks and ETFs might seem like a common approach in econometrics forecasting, the volatile nature of cryptocurrency suggests that it might be better to predict the realised variance (RV) to prevent extreme losses.

Our group report extensively analysed the daily RV of Ethereum (ETH). We used autoregressive (HAR and GARCH), tree-based (Random Forest), and deep-learning (LSTM) methods to predict the RV of ETH. We concluded that the two best methods to predict RV were HAR and Random Forest, as the predictions had the lowest root mean square error (RMSE) and highest R2.

The group report acknowledged that solely predicting the RV of ETH was not sufficient to capture the full dynamics of volatility within cryptocurrency. Given the high trading frequency, and volatile nature of cryptocurrency, this study aims to try and capture and predict the RV of different classification of cryptocurrencies.

This study improves on the group project with the following:

1. **Wider variety of coins:** Instead of solely forecasting ETH, this paper also wishes to forecast the RV of Bitcoin (BTC), Solana (SOL), Ripple (XRP), and Dogecoin (DOGE). It will be computationally expensive to create models for each cryptocurrency, and the cost might scale up when more cryptocurrencies are added. As such, this study will introduce a risk classification model to label each cryptocurrency.
2. **Expanding on models:** Our group project determined that autoregressive methods and bagging related models did well in predicting the RV of ETH, whereas deep learning methods such as LSTM did not perform well. I wish to expand on this finding and use more tree based models such as Bagging, and introduce the support vector regression (SVR) model to capture nonlinear dynamics.
3. **Different Train-Test split:** Dudek et al., 2021 used Feb 2017 to Dec 2018 as the training data, and then used Jan 2019 to Dec 2021 as the test data. In our group report, we used a 80/20 split which might have resulted in overfitting, given the volatile nature of cryptocurrency. This paper will align closer to Dudek et al.'s approach with respect to training and testing the models.

4. **Up to date data:** The data used in the group report were relatively outdated, with the last data point being 8 Oct 2023. This paper aims to use an updated dataset, with the latest data point being 10 Mar 2025.

Using training data between 1 April 2023 to 1 April 2024, this study produced 54 different models to forecast the RV of cryptocurrencies classified under different risk levels and for different frequencies.

This study determined that there were no one-size-fit-all models, as each risk level and frequencies had their own unique features and properties which suited certain models. However, this study determined that for hourly RV prediction, the GARCH model was the superior model, adapting well to the extremely noisy changes in RV. The simple exponentially weighted moving average model (EWMA) was able to predict the 3-hourly RV with the best accuracy. For the least noisy data, the daily RV, the bagging and SVR models did the best in forecasting the RV with decent accuracy.

## Methodology

The methodology follows [Dudek et al.'s \(2021\)](#) paper:

1. Training using data from 1 Apr 2023 to 1 Apr 2024:
  - a. Calculate the hourly, 3-hourly, daily, weekly, and monthly RV.
  - b. Use a composite score and classify the assets into three risk classifications: Low Risk, Medium Risk, and High Risk.
  - c. With respect to each risk classification, we generate 3 models to represent each forecast we intend to make, namely: hourly, 3-hourly, and daily forecasts
  - d. This will result in a total of nine models: three risk levels, three different frequencies for each risk level
2. Testing using data from 2 Apr 2024 to 10 Mar 2025:
  - a. Using the model generated for each frequency based on the training data, we will use a rolling window approach to generate a one-step, three-step, and 24-step ahead forecast to predict the hourly, 3-hourly, and daily  $\ln(RV)$  respectively.
  - b. Estimate the OOS RMSE and  $R^2$ , using a benchmark model as the base ratio.

## Train-test split

In our initial group report, we employed an 80/20 train-test split. While ETH's performance was favourable in terms of MSE and R2, the inherent noise in cryptocurrency RV raised concerns about overfitting. Following Dudek et al.'s approach, this study uses one year of data as the training data, and the subsequent year as the test set, resulting in a roughly balanced split (approximately 50/50, but 51/49 in reality).

- **Train:** 1 April 2023 - 1 April 2024
- **Test:** 2 April 2024 - 10 March 2024

This study compared the performance of the different splits using the baseline EWMA model. The 80/20 split produced lower MSE, but yielded higher R2 than a 50/50 split. Since the goal of the paper is to predict and explain the RV, this study decided to go with a 50/50 split. The findings are further explained in [Appendix 1](#).

## Dataset

The pricing of the cryptocurrencies are based on an hourly frequency, and obtained using the [Yahoo Finance API](#). Unlike the group project, which used minute-resolution data, this study used hourly-resolution data to strike a balance between data availability and computational efficiency. The key tradeoff here is that up-to-date minute-resolution data is only available for the period between 2016 and 2021, which limits the study's ability to study recent trends.

Another key difference between the group project and this study is that this study will not incorporate FRED-MD data. In our group project, we noticed that there was little to no correlation between the ETH's RV and FRED-MD data.

## Definition

Daily, weekly, and monthly RV are calculated in the same way as the group report. This study introduces hourly and 3-hourly RV.

| Table 1: Definition of RV |                                           |                                                                  |
|---------------------------|-------------------------------------------|------------------------------------------------------------------|
| Term                      | Description                               | Formula                                                          |
| <b>Hourly RV</b>          | Squared log returns of the hourly prices  | $RV_t^{(1H)} = (\ln P_t - \ln P_{t-1})^2$                        |
| <b>3-hourly RV</b>        | Sum of 3-hourly RV.                       | $RV_t^{(3H)} = \sum_{i=t-2}^t (\ln P_i - \ln P_{i-1})^2$         |
| <b>Daily RV</b>           | Sum of 24-hourly RV for cryptocurrencies. | $RV_t^{daily} = \sum_{i=0}^{23} RV_{t-i}^{hourly}$               |
| <b>Weekly RV</b>          | Mean of the sum of past 7-days of RV.     | $RV_t^{weekly} = \frac{1}{7} \sum_{i=0}^6 RV_{t-i}^{daily}$      |
| <b>Monthly RV</b>         | Mean of the sum of past 30-days of RV.    | $RV_t^{monthly} = \frac{1}{30} \sum_{i=0}^{29} RV_{t-i}^{daily}$ |

## Risk Level Classification

It is computationally expensive to train a separate model for each cryptocurrency, given that there exist 16,706 coins<sup>1</sup> in the market. As such, this study used a risk classification method to classify each coin into either Low, Medium, or High risk, and a model will be trained for each risk level. The classification is based on a composite scoring system, which places more weight on the long-term RV, and less weight on the short-term RV.

$$\text{Composite Score} = 0.2 \times \overline{\ln(RV_{\text{hourly}})} + 0.3 \times \overline{\ln(RV_{\text{3-hourly}})} + 0.5 \times \overline{\ln(RV_{\text{daily}})}$$

| Table 2: Coin Classification |            |                       |                         |                      |                    |
|------------------------------|------------|-----------------------|-------------------------|----------------------|--------------------|
| Coin                         | Risk       | Mean<br>ln(Hourly RV) | Mean ln(3<br>Hourly RV) | Mean<br>ln(Daily RV) | Composite<br>Score |
| <b>BTC/USD</b>               | <b>Low</b> | -13.0000              | -10.9511                | -8.21207             | -9.99134           |
| <b>ETH/USD</b>               |            | -12.7891              | -10.6885                | -7.94895             | -9.73885           |

<sup>1</sup> As of 12 March 2025, according to [CoinGecko](https://www.coingecko.com/)

|                 |             |          |          |          |          |
|-----------------|-------------|----------|----------|----------|----------|
| <b>XRP/USD</b>  | <b>Med</b>  | -12.2651 | -10.1707 | -7.45657 | -9.23251 |
| <b>DOGE/USD</b> | <b>High</b> | -12.1263 | -10.0259 | -7.29273 | -9.07939 |
| <b>SOL/USD</b>  |             | -11.2008 | -9.14405 | -6.52027 | -8.24351 |

## Exploratory Data Analysis

The exploratory data analytics was done using training data between 1 April 2023 to 31 March 2024.

### Summary Statistics

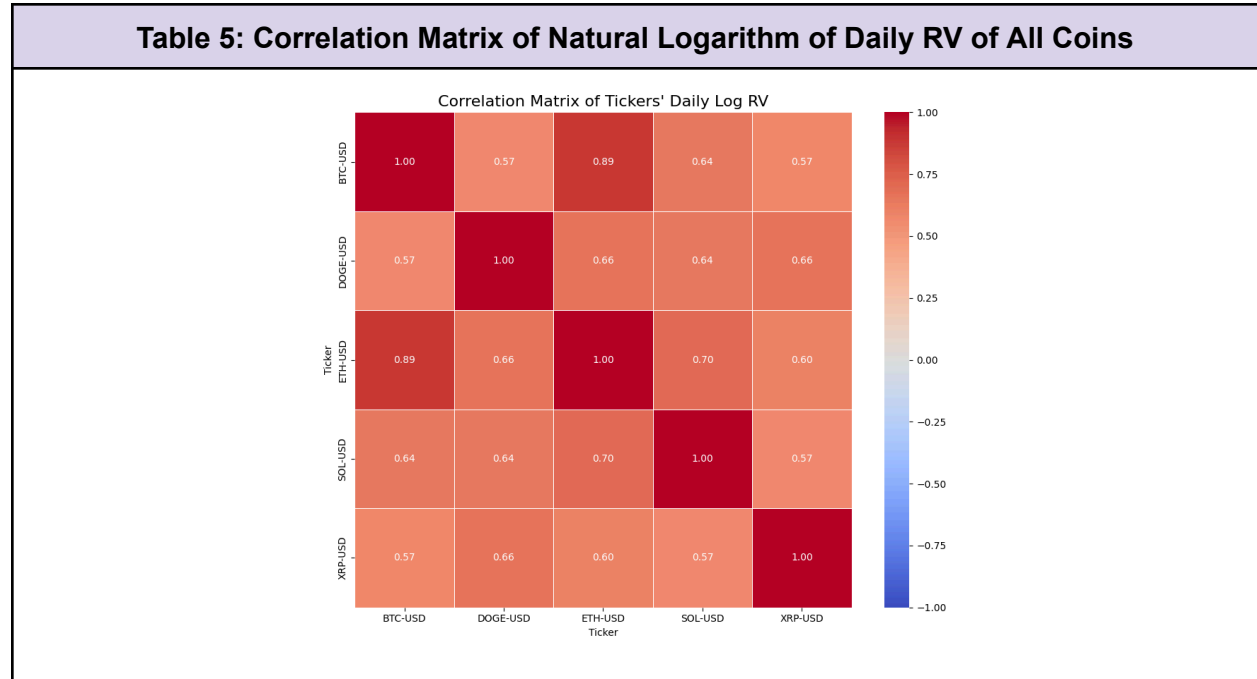
| Table 3: Natural Logarithm of Hourly Returns |             |          |          |           |          |
|----------------------------------------------|-------------|----------|----------|-----------|----------|
| Coin                                         | Risk        | Mean     | SD       | Min       | Max      |
| <b>BTC/USD</b>                               | <b>Low</b>  | 0.000104 | 0.004620 | -0.061683 | 0.053272 |
| <b>ETH/USD</b>                               |             | 0.000079 | 0.005164 | -0.073743 | 0.048796 |
| <b>XRP/USD</b>                               | <b>Med</b>  | 0.000019 | 0.007586 | -0.157708 | 0.218381 |
| <b>DOGE/USD</b>                              | <b>High</b> | 0.000120 | 0.009130 | -0.174408 | 0.174589 |
| <b>SOL/USD</b>                               |             | 0.000259 | 0.010164 | -0.152863 | 0.116354 |

All the cryptocurrencies had very low mean  $\ln(\text{Return})$ , but relatively high standard deviation.

| Table 4: Natural Logarithm of Hourly RV |             |          |          |          |          |
|-----------------------------------------|-------------|----------|----------|----------|----------|
| Coin                                    | Risk        | Mean     | SD       | Min      | Max      |
| <b>BTC/USD</b>                          | <b>Low</b>  | -13.0000 | 2.533356 | -27.7016 | -5.5715  |
| <b>ETH/USD</b>                          |             | -12.7891 | 2.588702 | -32.0984 | -5.21433 |
| <b>SOL/USD</b>                          | <b>Med</b>  | -12.2651 | 2.550954 | -26.43   | -3.04303 |
| <b>XRP/USD</b>                          | <b>High</b> | -12.1263 | 2.724969 | -31.8713 | -3.49064 |
| <b>DOGE/USD</b>                         |             | -11.2008 | 2.518743 | -26.5109 | -3.75643 |

Amongst the cryptocurrencies, the high risk coins had the smallest mean  $\ln(\text{RV})$ , and had relatively high SD as well.

## Correlation Matrix



All coins showed a positive correlation with each other, with ETH-BTC having the highest correlation, and ETH-SOL showing the second highest correlation.

## **Models**

This individual report will utilise two families of models: Autoregressive models (specifically HAR and GARCH) and Bagging models (specifically RF and Bagging with decision tree as the base model). Dudek et al. suggested the use of SVR-R, which produced favorable predictions for daily RV. Lastly, given the good performance of HAR in my group project, it would not be fair to use it as a baseline model. As such, this study will be using the exponentially weighted moving average (EWMA) as the new baseline model

### Baseline: Exponentially Weighted Moving Average Model

An exponentially weighted moving average (EWMA) model is mathematically simpler as compared to other models mentioned above. It is traditionally used in financial engineering as a baseline model, given the low computational cost and ease of interpretation. In an EWMA model, the variance is estimated using a weighted moving average.

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda)(r_{t-1} - \mu)^2$$

Where:

- $\lambda$  : Smoothing parameter between 0.9 to 0.99
- $r_{t-1}$  : Returns at t-1

[Bollen \(2014\)](#) suggested the optimal value of  $\lambda$  should be set to 0.97. However, it is noted that the study used the SPY index to obtain the above-mentioned  $\lambda$ . Given the volatile nature of cryptocurrency, it might not be suitable to assume the optimal  $\lambda$  to be 0.97. This study used the training data to obtain the optimal  $\lambda$ , which will be the  $\lambda$  value obtained that returned the lowest in-sample MSE.

| Table 6: Optimal EWMA $\lambda$ |            |                   |               |
|---------------------------------|------------|-------------------|---------------|
| Frequency                       | Risk Group | Optimal $\lambda$ | In-Sample MSE |
| Hourly                          | Low        | 0.915455          | 5.725802      |
|                                 | Medium     | 0.945455          | 6.099758      |
|                                 | High       | 0.962727          | 6.145357      |
| 3-Hourly                        | Low        | 0.9               | 1.649084      |
|                                 | Medium     | 0.9               | 1.680494      |
|                                 | High       | 0.9               | 1.721847      |
| Daily                           | Low        | 0.9               | 0.328084      |
|                                 | Medium     | 0.9               | 0.441546      |
|                                 | High       | 0.9               | 0.522555      |

The optimal  $\lambda$  fell between 0.90 to 0.96. A lower  $\lambda$  indicates that the model gives more weight to recent observations, making it more responsive to sudden price fluctuations, which is evident in the hourly data.

### HAR

HAR produced the best results when it came to predicting the daily RV for ETH, attaining the lowest MSE and highest  $R^2$ . In practice, HAR was extremely interpretable, and was computationally efficient. The formula of HAR is as such:

$$RV_t = \beta_0 + \beta_1 RV_{t-1} + \beta_2 RV_{t-3} + \beta_3 RV_{t-24} + \epsilon_t$$

To determine the coefficients to use to predict the RV, this paper [Corsi's \(2009\)](#) method with the following as the variables.:

- **Hourly:** Lagged hourly, Lagged 3-hourly, Lagged 24-hourly
- **3-Hourly:** Lagged 3-hourly, Lagged 12-hourly, Lagged 24-hourly
- **Daily:** Lagged daily, Lagged weekly, Lagged monthly.
  - This was what Corsi used to predict daily RV of stocks

I trained the model on the training data, and obtained the respective models for each risk level.

### GARCH(1,1)

Unlike the group project, this project did not tune the GARCH model given the large number of models to train (3 models for 3 frequencies). This study assumed that daily RV at time  $t$  is conditionally normally distributed, where  $F_{t-1}$  denotes the information set available at time  $t-1$ .

[Bollerslev \(1986\)](#) noted that  $\alpha_1 + \beta_1$  needs to be less than 1 to ensure stationarity. The GARCH model formula is as such:

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

Where:

- $\sigma_t^2$  is the conditional variance at time  $t$ ,
- $\omega$  is a constant (must be positive),
- $\alpha_1$  represents the coefficients of past squared residuals ( $\epsilon_{t-i}^2$ ) from the mean equation (ARCH terms),
- $\beta_1$  represents the coefficients of past variances ( $\sigma_{t-j}^2$ ) (GARCH terms),
- $p = 1$  is the number of lagged conditional variances (GARCH terms),
- $q = 1$  is the number of lagged squared residuals (ARCH terms),
- $\epsilon_t$  follows a white noise process.

This paper will be using the simplest possible GARCH model, which will be the GARCH(1,1) model. In a traditional GARCH(p,q) model,  $p$  represents the number of lagged conditional variances, and  $q$  represents the number of lagged squared residuals. Some papers tuned the GARCH model by finding the optimal  $p$  and  $q$  value which minimises the in-sample MSE. Given that there are 9 models to tune, it will be extremely computationally expensive to do so for each model.

| Table 7: GARCH Model Summary |            |          |          |          |          |
|------------------------------|------------|----------|----------|----------|----------|
| Frequency                    | Risk Group | Omega    | Alpha[1] | Beta[1]  | Mean     |
| Hourly                       | Low        | 0.829413 | 0.035758 | 0.836577 | -12.9073 |
|                              | Medium     | 0.321327 | 0.020011 | 0.932196 | -12.4559 |
|                              | High       | 0.06523  | 0.007049 | 0.983801 | -11.5492 |
| 3-Hourly                     | Low        | 0.673381 | 0.747441 | 0.000000 | -10.8    |
|                              | Medium     | 0.281431 | 0.338682 | 0.55842  | -10.339  |
|                              | High       | 0.291942 | 0.283287 | 0.619795 | -9.43998 |
| Daily                        | Low        | 0.026826 | 1.000000 | 9.80E-13 | -8.03715 |
|                              | Medium     | 0.005906 | 0.245224 | 0.754776 | -7.61552 |
|                              | High       | 0.004247 | 0.177884 | 0.822116 | -6.73693 |

### Random Forest

Random forest (RF) is an extremely popular method for forecasting financial indicators such as returns and volatility, since it is easy to implement and produces readable output. In our group project, RF produced one of the lowest MSE and highest R2. This study aims to replicate the performances of RF and achieve similar performances for other coins in different frequencies.

In the original group project, LASSO was used to pre-select the covariates prior to applying the RF. However, this study does not use any FRED-MD data, and as such, LASSO was not applied. Similar to GARCH, there will be no tuning of the RF model due to high computational cost. The model was implemented as follows:

```
model = RandomForestRegressor(n_estimators=100, random_state=42, max_depth=5)
```

### Bagging

Given the great performance of RF, the next natural step was to follow up with similar ensemble learning methods such as Bagging. Similar to RF, Bagging uses decision trees as the base models to predict the RV. But unlike RF, Bagging does not introduce random feature selection at

each split. Instead, it focuses solely on reducing overfitting through averaging predictions across multiple bootstrapped datasets.

## SVR

SVR is a model that was not previously used in the group project, but it will be introduced in this study. The reason for incorporating SVR is due to the ability to capture non-linear relationships present in the RV of different cryptocurrencies. By leveraging kernel functions, SVR should be able to model complex patterns in the data that traditional linear autoregressive methods may fail to capture.

## Results

### **Selection of Metrics**

This paper will be using four metrics to assess the respective models:

1. **Relative Root Mean Square Error:** The root mean square error (RMSE) measures the absolute difference between the predicted and actual value. This paper will be using the relative RMSE, with respect to the EWMA model as the benchmark.
  - a. *Relative RMSE = Model RMSE / EWMA RMSE* for each risk and time frequency
2. **R<sup>2</sup>:** R<sup>2</sup> measures how well the model explains the variances in predicting RV. A higher R<sup>2</sup> means that the model is able to predict the RV better, but also means it might be prone to overfitting
3. **Mean Absolute Percentage Error:** Mean absolute percentage error (MAPE) calculates the relative error between the predicted and actual value
4. **Quasi-Likelihood:** Quasi Likelihood (Q-Like) measures the goodness-of-fit of the model using a likelihood-based approach. Q-Like is useful when the likelihood function is hard to specify. A lower Q-Like value indicate that the model's predicted distribution closely matches the observed data

### **Model Results**

The full results of all the models can be found [here](#). Given that there are 54 models being produced (6 models for 3 different frequencies over 3 different risk levels), it will be hard to present the full results within this paper. As such, this paper has extracted the best performing models and presented it in the table below.

| Table 8: <a href="#">Best models</a> |            |              |               |          |          |          |
|--------------------------------------|------------|--------------|---------------|----------|----------|----------|
| Frequency                            | Risk Group | Model        | Relative RMSE | R2       | MAPE     | QLIKE    |
| Hourly                               | Low        | GARCH        | 0.996003      | 0.052877 | 0.153182 | 0.019054 |
|                                      | Medium     | GARCH        | 0.984106      | 0.164621 | 0.160922 | 0.021097 |
|                                      | High       | GARCH        | 0.948512      | 0.046368 | 0.16506  | 0.022216 |
| 3-Hourly                             | Low        | EWMA         | 1.000000      | 0.133228 | 0.115341 | 0.010146 |
|                                      | Medium     | EWMA         | 1.000000      | 0.368939 | 0.120725 | 0.011366 |
|                                      | High       | EWMA         | 1.000000      | 0.16083  | 0.117979 | 0.010765 |
| Daily                                | Low        | Bagging (DT) | 0.961873      | 0.151087 | 0.105051 | 0.008298 |
|                                      | Medium     | Bagging (DT) | 0.952226      | 0.388133 | 0.142263 | 0.014537 |
|                                      | High       | SVR          | 0.961328      | 0.158594 | 0.112303 | 0.009416 |

**Daily models had the lowest Q-Like and lowest absolute/relative RMSE as compared to hourly and 3-hourly frequencies.** This is because daily data had the least noise, with the data being relatively stationary across all time horizons, as compared to the extreme fluctuations observed in hourly and 3-hourly data. This was also further substantiated with the relatively high R2.

**GARCH was the best at predicting hourly RV across all risk groups.** This indicates that while other models struggled to capture the dynamics of hourly RV, GARCH did the best in predicting the RV. However, it is important to note that although GARCH attained the highest R2, it was still relatively low as compared to other timeframes, with the R2 ranging between 0.046 and 0.165. This is because hourly data tends to be noisier as compared to other timeframes, where it will be more susceptible to changes. The relative RMSE was extremely close to 1.000 as well, indicating that the EWMA model's performance might be extremely close to the GARCH model, and might serve as a good alternative to the GARCH model.

**Baseline EWMA model outperformed all other models in predicting the 3-hourly RV,** attaining the highest R2 and lowest RMSE. This shows that autoregressive and bagging models do not do a good job in predicting 3-hourly data. Despite being a baseline model, the EWMA model performed well in explaining the variance in predicting the 3 hourly RV.

**Bagging proved superior in predicting the low and medium risk daily RV, whereas the SVR did a better job in predicting the high risk daily RV.** Bagging models did well in predicting the daily RV, achieving a low relative RMSE, and had a high R2. SVR outperformed all other models in predicting the high risk coin as well, indicating that a kernel-based approach might be better in predicting daily RV for high risk coins

**The sole medium risk coin, XRP, achieved the highest R2 for all frequencies.** This was an interesting phenomenon, but more will be discussed in the discussion on why the behaviour of XRP was unique.

### **Diebold Mariano Test**

The Diebold-Mariano Test (DM test) is used to compare the forecasting accuracy between 2 models. While the DM test is used to test the predictive accuracy of two models, it cannot be used to definitively determine the suitability of the model. In this paper, the DM test will be used to test the predictive abilities between the models against the baseline EWMA model across all frequencies and risk levels. The findings of the DM test can be found [here](#).

The findings from the DM test substantiates the findings from the best model results obtained above, with the DM test results indicating that the EWMA model being favoured for the 3-hour trading intervals, and the GARCH and Bagging being favoured for the hourly and daily trading frequencies.

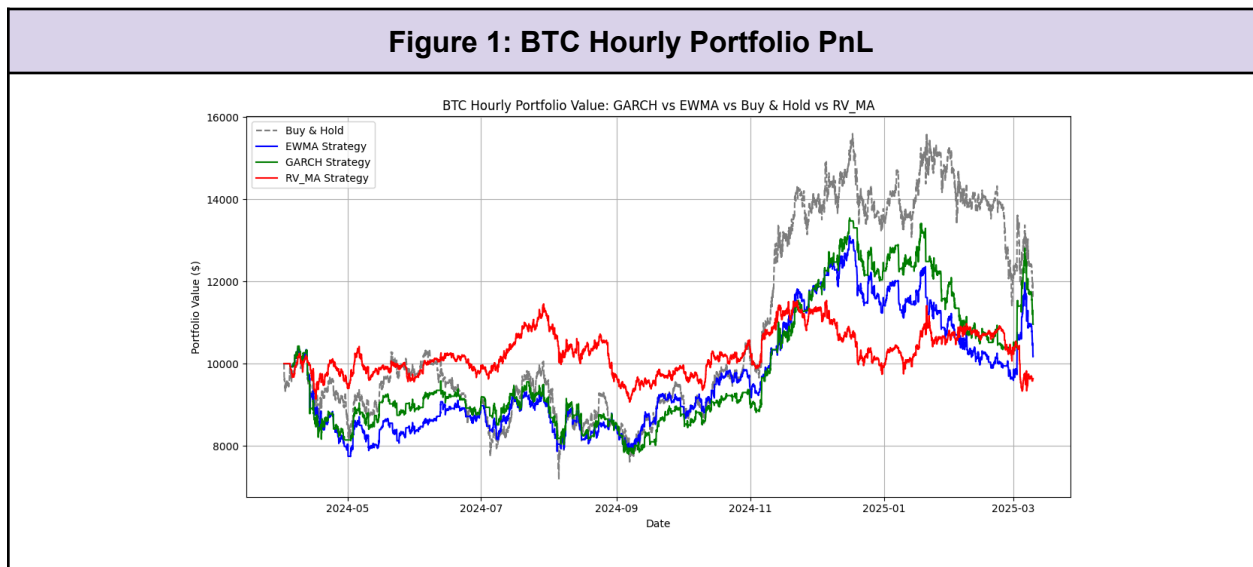
### **Application of Results: Naive Trading Portfolio**

In this section, this paper will apply a basic trading simulation using the trained models to evaluate the impact of forecasting RV and returns. This paper used coins from three different risk-classifications: BTC (low-risk), XRP (medium-risk), and DOGE (high-risk). The trading logic will be as such:

1. Begin with an initial capital of USD \$10,000

2. Initialise trading signals to get the median forecast of the past 3 days:
  - a. Hourly: Median forecast of the past 72 observations.
  - b. 3-Hourly: Median forecast of the past 24 observations.
  - c. Daily: Median forecast of the past 3 observations.
3. If forecast < signal, then will purchase the coin at the given price. Else, the model will not purchase the coin.
4. Sum up the cumulative returns over the test period.
5. Calculate the PnL.

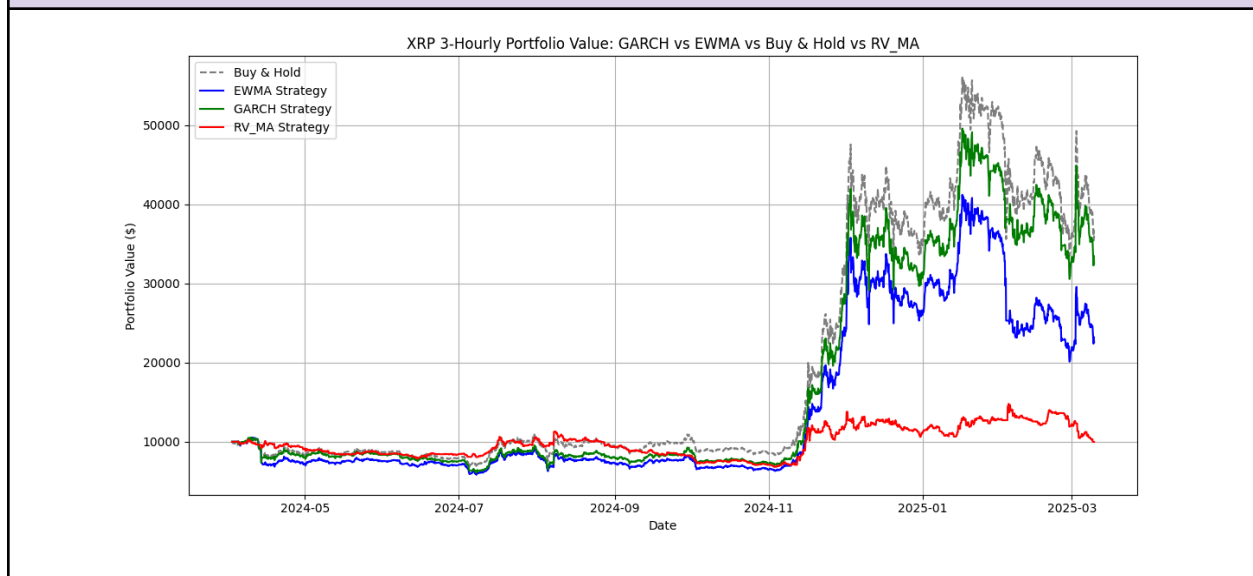
## Scenario 1: Hourly Trading With BTC



**In the hourly trading simulator, all models except for the RV moving average (RV MA) attained a profit.** The buy & hold model attained the highest profit, with \$1618, whereas the GARCH attained the second highest profit with \$956, and EWMA achieved \$177. RV MA attained a loss at \$366. Between April 2024 to October 2024, the RV MA model attained the highest profit. However, when the price of BTC started to increase, the buy and hold model was able to capture the highest profit, whereas the RV MA model made a loss during this period of high volatility.

## Scenario 2: 3-Hourly Trading With XRP

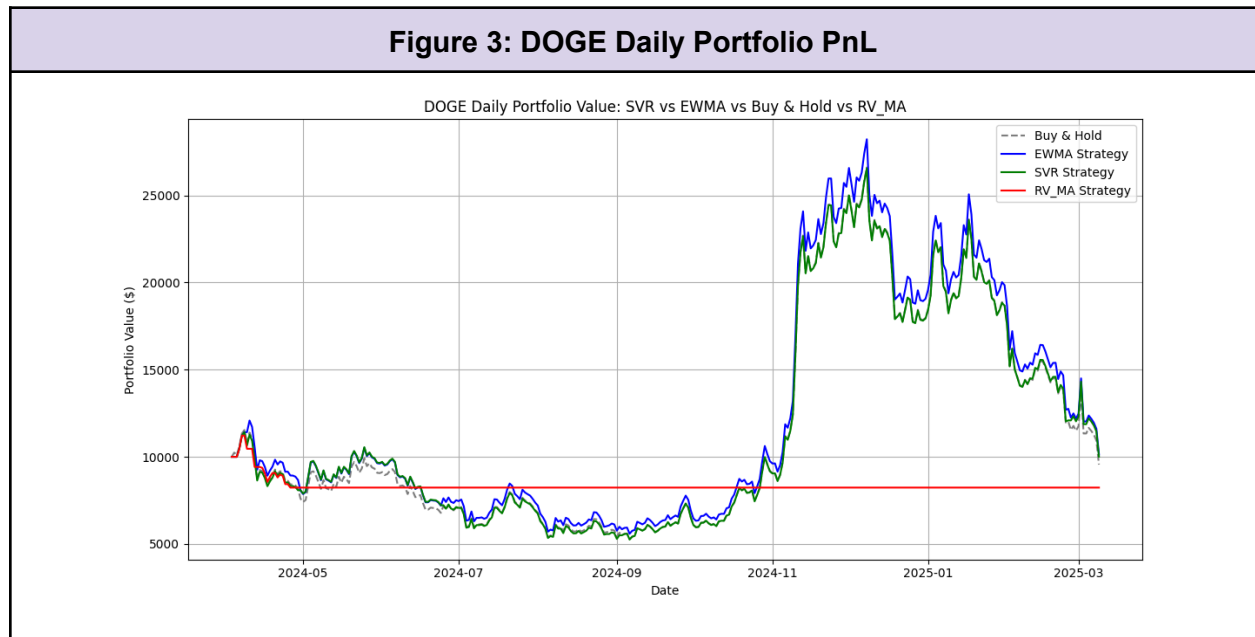
Figure 2: XRP 3-Hourly Portfolio PnL



**Despite having the best predictive ability, the EWMA model was unable to translate its forecasting ability into profits.** Across all models, the buy and hold model managed to obtain the highest profits at \$25696. GARCH once again came in second with \$22542 of profits and the EWMA strategy obtained a \$12577 profit. Once again, the RV MA model was the only model that recorded a loss of -\$26.

This paper acknowledges that the behaviour of XRP was unique, given that the price of XRP remained relatively constant until November 2024. This resulted in a good fit for the volatility prediction, and this was reflected in the models performing similarly to the buy-and-hold model until the boom in November 2024.

### Scenario 3: Daily Trading with Doge



Lastly, the EWMA model was able to obtain the highest profit for daily trades. But even so, the EWMA strategy was only able to attain a mere profit of \$137. **Even though the SVR was the best at predicting RV, it achieved a lower profit than EWMA.**

Interestingly, the RV MA model did not make any purchases after May 2024. This behaviour changed if a rolling median was used instead of a rolling mean, with more trades being performed.

### Discussion

Across all the frequencies and risk levels, this study noted that there is no one-size-fit-all solution when it comes to predicting the RV of cryptocurrencies. This underscores the complexity of cryptocurrency markets, and the nuanced nature of RV patterns. The key takeaway here was that although EWMA was used as a benchmark, most models scored quite close to the EWMA model, with the best models performing within 0.05 relative RMSE.

The test period of the data in this study also coincides with a very volatile global financial and political situation. Key financial and political events that happened between Apr 2024 to Mar 2025 includes: (1) 2024 United States Presidential Election, (2) Inauguration of Donald Trump,

(3) Announcement of U.S Crypto Strategic Reserve. Jeleskovic and Mackay (2023) shared that social media sentiment to news events, such as those mentioned above, plays a key role in predicting the pricing and volatility of cryptocurrency. Major events, such as the inauguration of President Donald Trump, will result in higher RV. This might also explain why the R2 value obtained in the group project was slightly higher than the R2 value obtained from this study.

## **GARCH Model**

The GARCH model obtained the highest R2, and lowest relative RMSE amongst all the models for hourly frequency. It is important to note the hourly RV for cryptocurrency is especially noisy, since the change in every hour will be more significant than a daily change. One unique property of GARCH is that it is able to adapt to heavy volatility shocks well, meaning that even when the RV changes due to external shocks such as news, it will be highly responsive to such shocks. Compared to the GARCH model, Bagging and RF will be prone to overfitting high frequency data, and HAR will be unable to react to sharp changes in the RV. Interestingly for the GARCH model, the low-risk hourly frequency had a significantly higher Omega value of 0.83, as compared to the high-risk hourly frequency of 0.06.

It is also noted that despite achieving the lowest relative RMSE and lowest R2, it barely outperformed the baseline model of EWMA. In addition, the GARCH model was computationally expensive to train, as a new model is fitted at every step to make a new prediction.

## **EWMA**

Despite not being a machine learning model, the econometric model EWMA produced the best R2 for the 3-hourly RV forecast, with the lowest relative RMSE (of 1, since it is already the baseline model). This paper found out that the optimal lambda range was less than the 0.97 suggested by Bollin. This means that the EWMA model was trained to be much more responsive, which follows the literature of cryptocurrency being more volatile and having shorter memory than traditional assets like SPY. Even though the EWMA model was trained to be more responsive, it was still unable to capture the full volatility extent of the hourly RV. Nonetheless, across different frequencies, the EWMA model achieved competitive Q-Like and MAPE values, meaning that it is a very reliable baseline model which can be deemed as an alternate model.

## Bagging Model vs RF

Given the outstanding performance of the RF model in the group project, this paper employed the bagging model. In theory, the bagging model is more prone to overfitting than RF, since the RF model adds a layer of randomness by selecting a random subset of predictors at each split. [Breiman \(2001\)](#) paper on RF expected the RF model to outperform the bagging model in most cases because of this additional layer on randomness. However, this paper found out that bagging marginally outperforms RF. These findings align with [Altman and Krzywinski \(2017\)](#), which stated that although RF is objectively a better prediction model, the extra randomness of RF may sometimes dilute the influence of very strong predictors.

Another possible reason for the differing performance was that some features, such as lagged predictors, outperformed some features that the RF sampled out. This means that using all features, like how it is done in bagging, might have resulted in higher R2 and lower relative RMSE.

## HAR

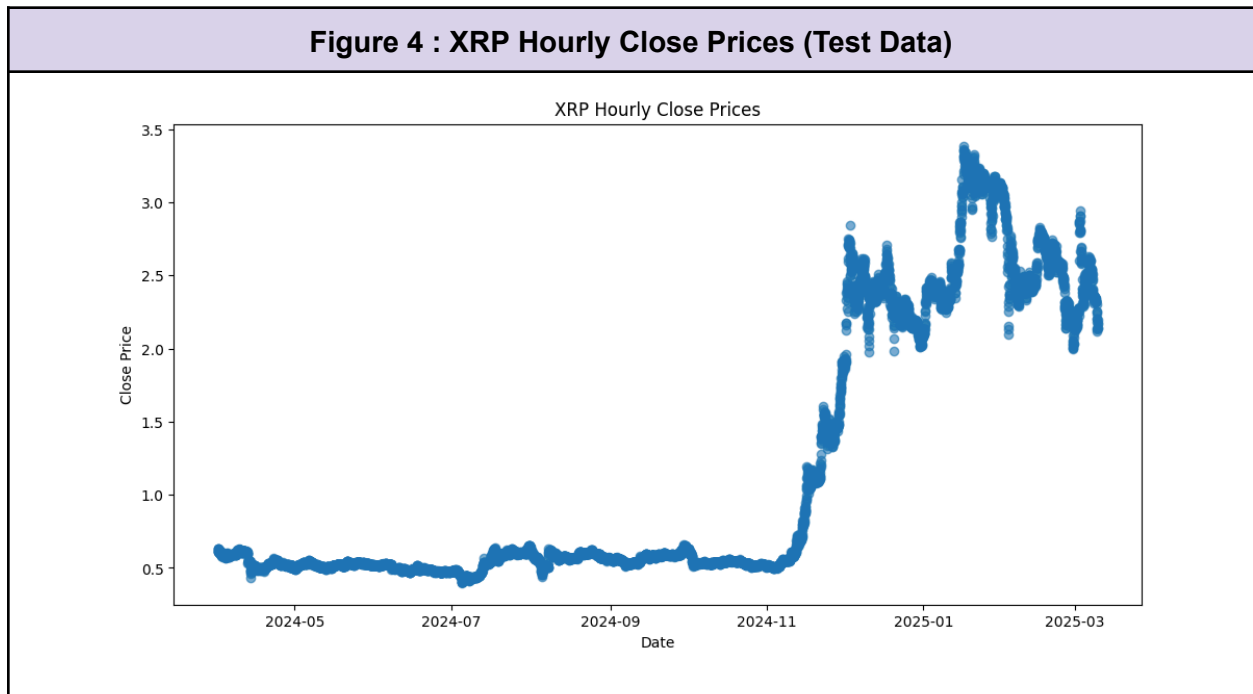
Despite being a baseline model in the group project, HAR produced the best forecast when predicting the daily RV of ETH. In this paper, the HAR model severely underperformed against all other models across all time periods.

One possible reason why HAR underperformed might be because it does not have short-term memory, and might be more suitable for long-term memory models. However, the results obtained proved otherwise, as the HAR model performed the worst amongst all models in the daily prediction, and it actually performed relatively well in the hourly and 3-hourly predictions.

## SVR

The SVR model was a new model introduced in this paper. Unlike GARCH and HAR, SVR uses a kernel-based approach, meaning that it maps features into higher dimensional spaces. In this paper, the SVR model had the best predictive power for high-risk daily RV prediction, but it was outperformed by other models for other use cases. In fact, the SVR model was unable to yield the highest simulated trading return, losing out to the baseline EWMA model.

## The Caveat of Ripple (XRP) Coin



From Figure 4, this study notes that XRP has a unique trend with respect to the hourly closing price from test data (Apr 2024 - Mar 2025). Between Apr 2024 to Nov 2024, XRP's closing price did not change much, remaining at approximately 0.5. This resulted in better performance of the medium risk coin's prediction. Based on the pattern of the closing price between Nov 2024 to Mar 2025, it can even be assumed that the XRP coin will most probably be classified as a high risk coin.

### Good RV Predictive Ability Does Not Translate To Profits

In the 3 simulated trading scenarios, the models with the highest  $R^2$  and lowest MSE were unable to achieve the highest profit. In fact, the buy and hold model often outperformed the performance of the more complex models such as EWMA, GARCH, and SVR. However, all models outperformed the most basic RV MA model, meaning that if the trader intends to use the moving median of the RV, it will not outperform the more complex models, but a buy-and-hold model might be the most appropriate trading strategy for cryptocurrency.

Buy-and-hold models might be more effective for cryptocurrency as it captures the full trend of the market, meaning that in times of a bearish market like the test data, it would be able to reap

the full benefits and profits of the market. However, in a more bearish or volatile timeframe, such as Liberation Day (April 4, 2025), the buy-and-hold model might suffer from the sharp swings within the market.

Lastly, it is important to note that the RV solely forecasts the risk and magnitude of price movements; it does not forecast the direction in which the asset will move. This means that the current trading strategy based solely on low predicted RV might result in very undesirable results. There are much more complex trading models that uses the volatility as a trading indicator, more will be discussed in [Future Works](#)

## Conclusion

This paper examines the dynamics between the type of cryptocurrency and the predictive ability of different models. This paper expanded on the previous findings which focused solely on predicting the RV of ETH by analysing a diverse set of cryptocurrencies across multiple frequencies. **The core finding underscores the absence of a single superior model for forecasting RV across different frequencies and risk levels.** However, this paper managed to identify that for predicting hourly RV, GARCH(1,1) seemed to be the best, for 3-hourly RV, the EWMA was the best, and for daily RV, the bagging and SVR model performed the best.

The most surprising finding of this paper was the positive performance of the newly added baseline EWMA model. As mentioned in the methodology, the EWMA model was even simpler than the previous baseline model of HAR, but in tougher scenarios, the EWMA model managed to outperform ML models such as RF, Bagging, and SVR. On the contrary, the HAR underperformed significantly in this paper. Even in scenarios where models outperformed the EWMA model, it was deemed as marginal, with the relative difference being less than 0.1.

In the application section of this paper, the model which has the most accurate RV forecast might not have the highest profit when it comes to a RV-signalled trading algorithm. Oftentimes, the buy-and-hold model outperforms the more sophisticated models, albeit by a small margin. This disconnect in findings emphasises that although volatility forecasting can offer valuable insights to risk management, it might not be the most suitable trading strategy for cryptocurrencies.

## Future Works

[Gyamerah \(2019\)](#) worked on predicting the returns of Bitcoin using different variants of the GARCH model, concluding that the tGARCH-NIG model best captured the daily Bitcoin volatility. While this paper found that GARCH models were able to produce high R2 and low relative RMSE, it still has space for tuning and improvement. One potential area of improvement is to see how different cryptocurrencies interact with the tGARCH-NIG model. Another model not explored, due to time constraints, was the XGBoost model, which is expected to produce similar results to the other types of bagging models.

**Given the sentiment-driven performance of cryptocurrency, the logical next step will be to use news headlines to further improve on the prediction of RV.** Combining news headlines with methods used in this paper such as GARCH and HAR to predict the RV of assets has been done by [Bodilson and Lunde \(2024\)](#). Their paper noted that using news headlines improved the forecast accuracy for long-horizon prediction, and this finding serves as a natural springboard for the next phase of the project.

**Another area of improvement identified would have been the need to tune the models, specifically bagging, RF, and SVR.** This paper has attempted to tune the above-mentioned models, but was unable to do so due to the intensive computational demands to tune 9 different models per ML model. Hyperparameter tuning involves using a searching algorithm, like grid-search, to identify the optimal combination of model parameters that minimises the prediction error in the validation data. For instance, when optimising the RF model, this paper could use optimal tuning parameters such as `n_estimators` and `max_depth` to obtain the lowest in-sample MSE. Further research should be done to do hyperparameter tuning to improve the accuracy of the models.

The application of results in a trading method demonstrated that even with the best RV prediction model, profits might not be guaranteed. **However, it is acknowledged that the trading model used was overly simplified, with the only trading signal used was the rolling median of the RV.** [Moreira and Muir \(2017\)](#) developed the 'Volatility-Managed Portfolio'. Unlike the simplified model used in this paper, the Volatility-Managed Portfolio aims to maximise the Sharpe Ratio by dynamically adjusting the allocation to a risky asset based on its predicted volatility. So instead of just buying and selling one asset, it considers the interactions between

multiple assets, and will buy and sell assets according to the predicted RV of each portfolio. While not perfect, this model will serve as a step up from the current basic model used.

## Appendix

### **Appendix 1: Benefits of using a 50/50 train-test split, compared to an 80/20 train-test split**

This appendix analyses the difference between a 50/50 and 80/20 train-test split. To make a simplified comparison, this section will be using the baseline EWMA model, and comparing the hourly forecast. The reason for selecting the hourly forecast is because the hourly data is considered the “noisiest”, so we will be able to examine how the EWMA model will react to the noisy data under different splits.

The EWMA models were trained similarly, with the only difference being the train-test splits. The difference is calculated by taking the difference between the 80/20 and 50/50. The findings are as such:

| Table A1: Train-Test Split Comparison |            |          |                |               |
|---------------------------------------|------------|----------|----------------|---------------|
| Frequency                             | Risk Group | Ticker   | MSE Difference | R2 Difference |
| Hourly                                | Low        | ETH-USD  | -0.38838       | -0.01416      |
|                                       |            | BTC-USD  | -0.33465       | -0.00997      |
|                                       | Medium     | XRP-USD  | -0.09523       | -0.01058      |
|                                       | High       | SOL-USD  | -0.23788       | 0.003418      |
|                                       |            | DOGE-USD | 0.087419       | -0.01574      |

The MSE difference across most tickers was negative, with the exception being DOGE, which had a difference that is close to zero. **This means that the MSE of the 80/20 was smaller than the MSE of the 50/50 split.** On the other hand, **the R2 for the 80/20 was smaller than the R2 of the 50/50 split.** This means that predictive abilities of the 50/50 split was better than the 80/20 split.

In this study, there is a discrepancy between the MSE and  $R^2$ , which makes the selection between a 50/50 and 80/20 split more challenging. This boils down to the purpose of the study: The goal is to determine which model is the best in predicting the RV, and this means we will be comparing the  $R^2$ . In fact, a low MSE might suggest that there might be overfitting, and as such, this study decides to use a 50/50 split instead.

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